WORKED SOLUTION

STEP I 2017, Question 7

Pure • Napoleon's Theorem

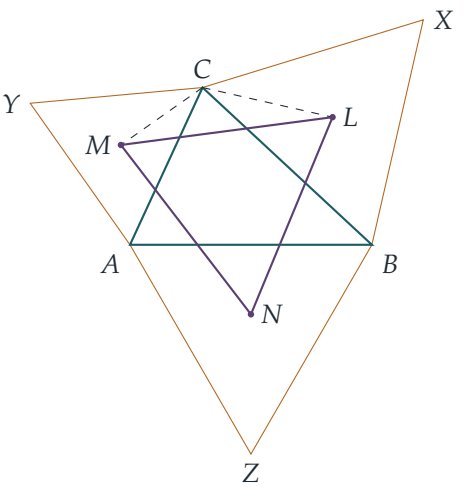
STEP I 2017, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s1-q7

The key idea

Everything happens at the vertices of ABC. The centre of an equilateral triangle sits on the bisector of each of its angles, so the lines from C to L and to M each make $30°$ with a side of ABC. That gives the angle LCM in terms of C. The cosine rule in triangle LCM then gives $|LM|$. The answer turns out to be symmetric in a, b, c, and that symmetry is what makes LMN equilateral.



Solution

Part (i)

In the equilateral triangle CYA of side b, the centre M lies on all three medians, which are also the altitudes and angle bisectors. The median from C has length $\frac{\sqrt{3}}{2}b$, and the

centre is two-thirds of the way along it from the vertex. So

$$|CM| = \tfrac{2}{3} \cdot \tfrac{\sqrt{3}}{2}\, b = \frac{b}{\sqrt{3}}.$$

In the same way, triangle BXC has side a, so

$$|CL| = \frac{a}{\sqrt{3}}.$$

Part (ii)

Since CM bisects the 60° angle YCA, the line CM makes 30° with CA, on the side away from B. Likewise CL makes 30° with CB, on the side away from A. Going round C from L to M we pass through 30°, then the angle C of the triangle, then another 30°. So

$$\angle LCM = C + 60^\circ.$$

? **Ask yourself** What if $C > 120^\circ$, so that $C + 60^\circ$ is more than 180°?

Then the angle of triangle LCM at C is the other way round, $360^\circ - (C + 60^\circ)$. But $\cos(360^\circ - \theta) = \cos\theta$, so the cosine rule below gives the same result in every case.

Applying the cosine rule in triangle LCM, with the lengths from part (i):

$$\begin{aligned} |LM|^2 &= \frac{a^2}{3} + \frac{b^2}{3} - \frac{2ab}{3}\cos(C + 60^\circ) \\ &= \frac{a^2 + b^2}{3} - \frac{2ab}{3}\left(\tfrac{1}{2}\cos C - \tfrac{\sqrt{3}}{2}\sin C\right) \\ &= \frac{a^2 + b^2}{3} - \frac{ab\cos C}{3} + \frac{\sqrt{3}\,ab\sin C}{3}. \end{aligned}$$

Now we use two facts about triangle ABC. The cosine rule gives $2ab\cos C = a^2 + b^2 - c^2$, and the area is $\Delta = \frac{1}{2}ab\sin C$, so $ab\sin C = 2\Delta$. Multiplying by 6,

$$\begin{aligned} 6|LM|^2 &= 2(a^2 + b^2) - 2ab\cos C + 2\sqrt{3}\,ab\sin C \\ &= 2a^2 + 2b^2 - (a^2 + b^2 - c^2) + 4\sqrt{3}\,\Delta \\ &= a^2 + b^2 + c^2 + 4\sqrt{3}\,\Delta, \end{aligned}$$

as required.

The right-hand side is unchanged if we permute a, b, c, and Δ does not depend on how we label the triangle. Repeating the argument at vertex A (for $|MN|$) and at vertex B (for $|NL|$) gives the same expression. So

$$|LM| = |MN| = |NL|,$$

and LMN is equilateral.

Equal areas. An equilateral triangle of side s has area $\frac{\sqrt{3}}{4}s^2$. Write $S = a^2 + b^2 + c^2$. Then

$$[LMN] = \frac{\sqrt{3}}{4}|LM|^2 = \frac{\sqrt{3}}{24}(S + 4\sqrt{3}\,\Delta) = \frac{\sqrt{3}}{24}S + \frac{\Delta}{2}.$$

Hence

$$[LMN] = \Delta \iff \frac{\sqrt{3}}{24}S = \frac{\Delta}{2} \iff S = \frac{12\Delta}{\sqrt{3}} = 4\sqrt{3}\,\Delta.$$

Every step is reversible, so this is the required "if and only if".

Part (iii)

Write everything in terms of a, b and C, using $c^2 = a^2 + b^2 - 2ab\cos C$ and $\Delta = \frac{1}{2}ab\sin C$. Each line below is equivalent to the one before:

$$\begin{aligned} & a^2 + b^2 + c^2 = 4\sqrt{3}\,\Delta \\ \iff \quad & 2a^2 + 2b^2 - 2ab\cos C = 2\sqrt{3}\,ab\sin C \\ \iff \quad & a^2 + b^2 = ab(\cos C + \sqrt{3}\sin C) \\ \iff \quad & a^2 + b^2 = 2ab\cos(C - 60^\circ), \end{aligned}$$

because $\cos C + \sqrt{3}\sin C = 2(\cos C\cos 60^\circ + \sin C\sin 60^\circ) = 2\cos(C - 60^\circ)$. Subtracting $2ab$ from both sides,

$$a^2 - 2ab + b^2 = 2ab\cos(C - 60^\circ) - 2ab \iff (a - b)^2 = -2ab(1 - \cos(C - 60^\circ)).$$

So the two conditions are equivalent.

> **?** **Ask yourself** How can a square equal something with a minus sign in front?

Only if both sides are zero. The left-hand side $(a - b)^2$ is at least 0. On the right, $ab > 0$ and $1 - \cos(C - 60^\circ) \geq 0$, so the right-hand side is at most 0. Equality therefore holds exactly when

$$a = b \quad \text{and} \quad \cos(C - 60^\circ) = 1.$$

Since $0^\circ < C < 180^\circ$, we have $-60^\circ < C - 60^\circ < 120^\circ$, and the only angle in this range with cosine 1 is 0°. So $C = 60^\circ$. A triangle with $a = b$ and an apex angle of 60° has its other two angles equal, summing to 120°, so all three angles are 60° and the triangle is equilateral. Conversely, an equilateral triangle has $a = b$ and $C = 60^\circ$, so it satisfies the condition.

Combining this with part (ii): $[LMN] = [ABC] \iff a^2 + b^2 + c^2 = 4\sqrt{3}\,\Delta \iff ABC$ is equilateral.

Checks

- Equilateral ABC with side 1: then $\Delta = \frac{\sqrt{3}}{4}$ and $6|LM|^2 = 3 + 3 = 6$, so $|LM| = 1$, and the areas are equal, as part (iii) predicts.

- The 3–4–5 right-angled triangle: $S = 50$ and $\Delta = 6$, so $|LM|^2 = (50 + 24\sqrt{3})/6 \approx 15.26$. Then $[LMN] \approx 6.61$, which is bigger than $[ABC] = 6$. A computer check with coordinates for the points L, M, N gave the same values, and confirmed the formula for $6|LM|^2$ on thousands of random triangles, including obtuse ones.
- Part (iii) also shows $a^2 + b^2 + c^2 \geq 4\sqrt{3}\,\Delta$ for every triangle, so $[LMN] \geq [ABC]$ always.

Common slips

- Taking $|CM|$ to be the whole median $\frac{\sqrt{3}}{2}b$, or one-third of it, instead of two-thirds.
- Using $\angle LCM = C + 30^\circ$ or C, forgetting the 30° on each side.
- In part (iii), stopping at "$a = b$". You also need $C = 60^\circ$, and the range of C to rule out other angles with cosine 1.

Beyond STEP

This is Napoleon's theorem: the centres of equilateral triangles erected outwards on the sides of any triangle form an equilateral triangle. The inequality $a^2 + b^2 + c^2 \geq 4\sqrt{3}\,\Delta$ found in part (iii) is Weitzenböck's inequality. Complex numbers give a very slick proof of Napoleon's theorem: rotating by 60° is multiplication by $e^{i\pi/3}$. Needham's *Visual Complex Analysis* develops this way of doing geometry beautifully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.